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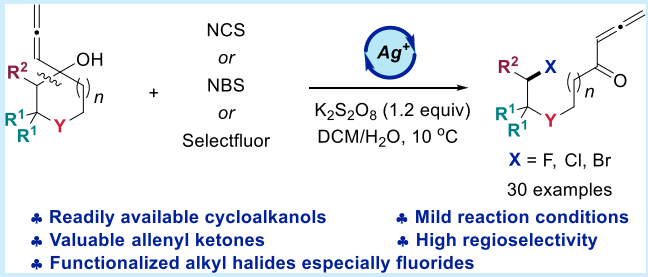
Halogen-Substituted Allenyl Ketones through Ring Opening of Nonstrained Cycloalkanols

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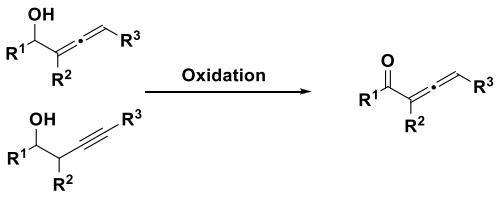
ABSTRACT: An efficient synthesis of halogen-substituted allenyl ketones via Ag-catalyzed oxidative ring opening of allenyl cyclic alcohols under mild reaction conditions has been achieved. The reaction features a wide substrate scope and excellent regioselectivity. The synthetic potential of the products has been demonstrated by their conversion to stereodefined alkenes and heterocyclic compounds.



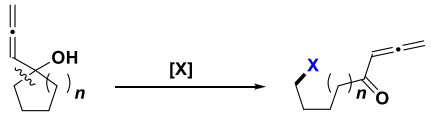
Allenes are an important class of molecules because of their unique cumulated double bonds, which are found in many bioactive natural products and pharmaceuticals.[1] Among various allene derivatives, allenyl ketones are excellent electrophiles that undergo nucleophilic addition and cyclization reactions.[2] Consequently, much attention has been paid to synthetic routes for allenyl ketones.[3] Oxidations of secondary 2,3-allenols and homopropargylic alcohols are the most straightforward approaches (Scheme 1a).[4] On the other hand, much attention has recently been paid to the catalytic C−C bond cleavage of nonstrained systems.[5,6] Here we wish to report the oxidative ring-opening reaction of allenyl cyclic alcohols to afford halogen-substituted allenyl ketones (Scheme 1b).

Scheme 1. Oxidation of Alcohols to Allenyl Ketones

a) The oxidation of secondary alcohols



b) The oxidative ring-opening of tertiary allenyl cycloalkanols (this work)



In the beginning, we used 1-propadienylcyclohexanol (**1a**) as the model substrate and NCS (**2a**) as the chlorine source to optimize the reaction conditions. First, different mixed solvents were screened. THF/H_2O resulted in no product **3aa** (Table 1, entry 1). Other mixed solvents such as MeCN/H_2O, EA/H_2O, and *c*-hexane/H_2O provided the desired product **3aa** in 5−32% yield (Table 1, entries 2−4). It was observed that toluene/H_2O was the best mixed solvent and that DCM/H_2O had almost the same effect (58−60% yields; Table 1, entries 5 and 6). We further reduced the amount of $AgNO_3$, and 20.0 mol % was enough for the reaction (Table 1, entry 7). Then we tried different silver catalysts such as $AgSbF_6$, $AgPF_6$, $AgBF_4$, and AgOTs. All of the catalysts provided lower yields of the desired product **3aa** (26−40%; Table 1, entries 8−11). Subsequently, we also tested some other oxidants such as TBHP and $Na_2S_2O_8$ and found that $K_2S_2O_8$ was still the best oxidant (Table 1, entries 12 and 13). Next, we observed that lowering the temperature to 10 °C increased the yield of **3aa** to 65% with recovery of 9% of starting material **1a** (Table 1, entry 14). Decreasing the temperature further to 0 °C did not improve the yield of **3aa** (60%; Table 1, entry 15), and increasing the loading of $AgNO_3$ to 24.0 mol % at 10 °C led to a 67% yield of product **3aa** with trace recovery of **1a** (Table 1, entry 16). We also conducted the reaction in the air atmosphere, and the NMR yield of product **3aa** dramatically decreased to 27% (Table 1, entry 17). Finally, the yield of **3aa** was further improved to 77% when the reaction was carried out on a 1.0 mmol scale (Table 1, entry 18).

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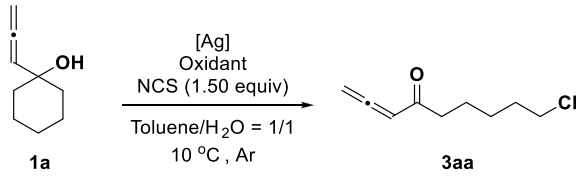


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Table 1. Optimization of the Reaction Conditions[a]



1a 3aa

entry	[Ag]	oxidant	solvent	T (°C)	yields of 3aa/1a (%)[b]
1	$AgNO_3$	$K_2S_2O_8$	THF	25	0/0
2	$AgNO_3$	$K_2S_2O_8$	MeCN	25	5/0
3	$AgNO_3$	$K_2S_2O_8$	EA	25	27/0
4	$AgNO_3$	$K_2S_2O_8$	*c*-hexane	25	32/0
5	$AgNO_3$	$K_2S_2O_8$	DCM	25	58/0
6	$AgNO_3$	$K_2S_2O_8$	toluene	25	60/0
7[c]	$AgNO_3$	$K_2S_2O_8$	toluene	25	58/0
8[c]	$AgSbF_6$	$K_2S_2O_8$	toluene	25	26/6
9[c]	$AgPF_6$	$K_2S_2O_8$	toluene	25	34/0
10[c]	$AgBF_4$	$K_2S_2O_8$	toluene	25	37/0
11[c]	AgOTs	$K_2S_2O_8$	toluene	25	40/0
12[c]	$AgNO_3$	TBHP	toluene	25	0/85
13[c]	$AgNO_3$	$Na_2S_2O_8$	toluene	25	38/6
14[c,d]	$AgNO_3$	$K_2S_2O_8$	toluene	10	65/9
15[c,d]	$AgNO_3$	$K_2S_2O_8$	toluene	0	60/14
16[d,e]	$AgNO_3$	$K_2S_2O_8$	toluene	10	67/trace
17[d,e,f]	$AgNO_3$	$K_2S_2O_8$	toluene	10	27/48
18[d,e,g]	$AgNO_3$	$K_2S_2O_8$	toluene	10	77/2

[a]The reaction was conducted with 0.2 mmol of **1a**, 0.5 equiv of $AgNO_3$, 2.5 equiv of $K_2S_2O_8$, and 1.5 equiv of NCS in 0.5 mL of H_2O and 0.5 mL of toluene at 25 °C for 24 h under an Ar atmosphere. [b]NMR yields with CH_2Br_2 as the internal standard. [c]0.20 equiv of $AgNO_3$ and 1.0 equiv of $K_2S_2O_8$ were applied. [d]The reaction time was 12 h. [e]0.24 equiv of $AgNO_3$ and 1.2 of equiv $K_2S_2O_8$ were applied. [f]The reaction was conducted in air. [g]1.0 mmol scale.

With the optimized reaction conditions in hand, the substrate scope of various nonstrained cycloalkanols **1** with different halogen sources was investigated (Figure 1). First of all, we evaluated the ring sizes of cycloalkanols with different halogen sources. We found that when NCS was used as the halogen source, cycloalkanols with ring sizes of 5−8 could be applied, generating **3aa**−**da** smoothly. NBS and even Selectfluor could be used as the halogen source to afford the corresponding ketones in moderate yields (**3ab**−**cb**, **3ac**, **3cc**, and **3dc**). Because the C−F bond in saturated systems shows low surface tension,[7] increased lipophilicity of bioavailability,[8] high thermal stability,[9] and isoelectronic effects to −OH, $-NH_2$ and $-CH_3$,[10] the synthesis of alkyl fluorides is of high current interest.[11] Here we efficiently prepared potentially useful alkyl fluorides **3ac**, **3cc**, **3dc**, and **3kc** with an allenyl ketone unit, which are difficult to prepare by known methods. Next, reactions using a wide range of cyclohexanols substituted with methyl, ethyl, dimethyl, carbonyl, and acetal at the 4-position led to the formation of the desired products **3ea**−**ia** and **3eb**−**ib**. Besides, the reactions of 4-allenyltetrahydro-2*H*-pyran-4-ol also delivered the desired allene products **3ja** and **3jb** in 30 and 40% yield, respectively.

Furthermore, with nonsymmetrical examples, excellent selectivity was observed for the C−C bond cleavage in the α-substituted substrates, affording the products **3ka**, **3kb**, **3kc**, and **3lb** in decent yields exclusively. High regioselectivity was observed for the bridged bicyclic substrate **1m**, affording **3ma** and **3mb** as the main products (**3ma**:**3ma′** = 96:4, **3mb**:**3mb′**

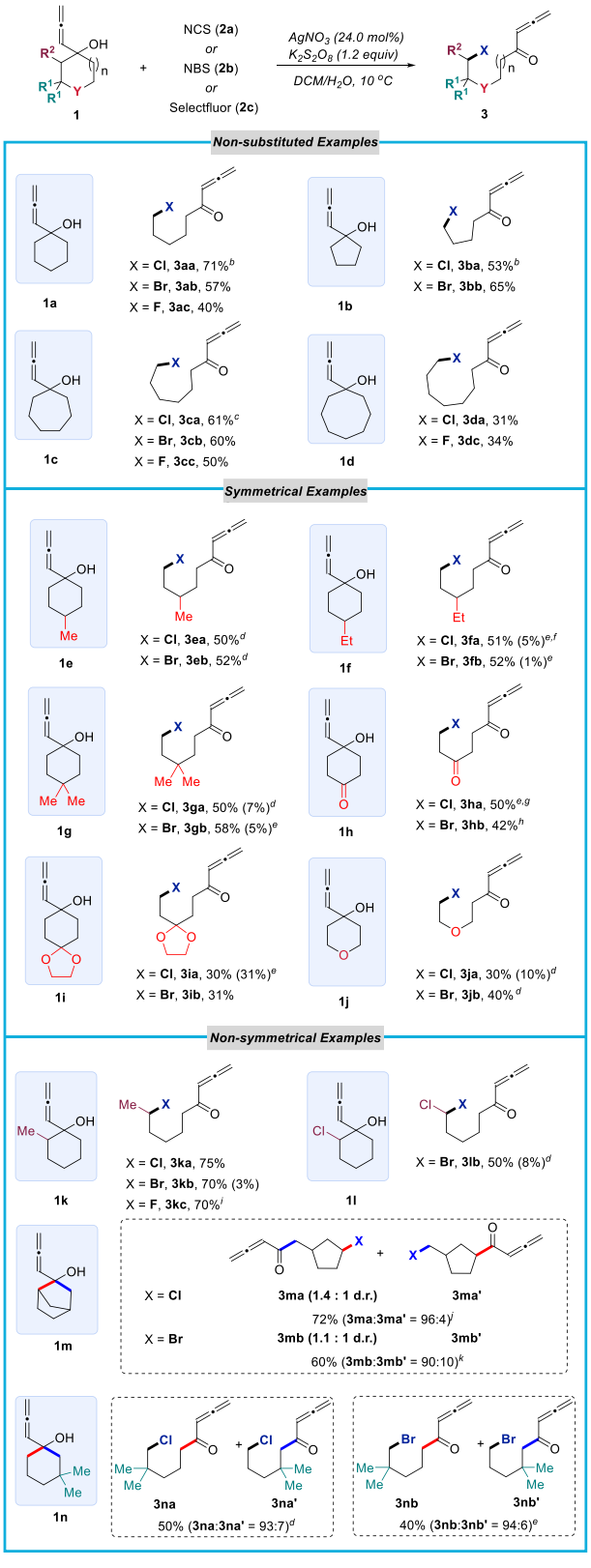


Figure 1. Substrate scope. Notes: [a]The reaction was conducted with 1.0 mmol of **1**, 0.24 equiv of $AgNO_3$, 1.20 equiv of $K_2S_2O_8$, and 1.5 equiv of NCS (**2a**) or NBS (**2b**) or 2.0 equiv of Selectfluor (**2c**) in 2.5 mL of H_2O and 2.5 mL of DCM at 10 °C for 12 h under an Ar atmosphere. Numbers in parentheses are the yields of the recovered starting cyclohexanols **1**. [b]2.5 mL of toluene was used instead of 2.5 mL of DCM. [c]With 0.40 equiv of $AgNO_3$ and 2.0 equiv of $K_2S_2O_8$. [d]With 0.50 equiv of $AgNO_3$ and 2.5 equiv of $K_2S_2O_8$. [e]With 0.50 equiv

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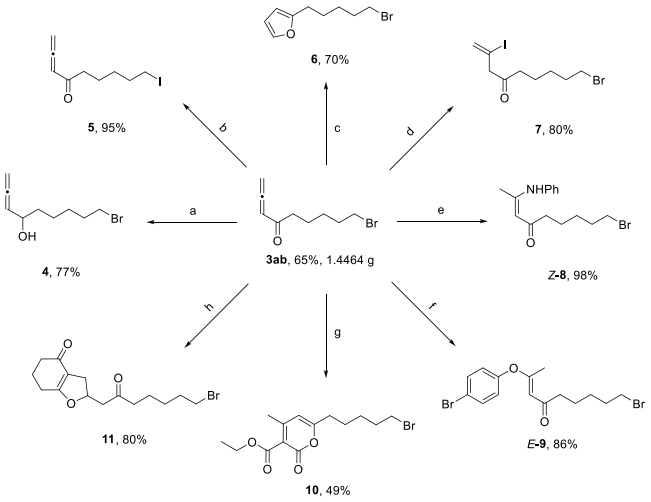
Figure 1. continued

of $AgNO_3$. [f]The reaction was run on a 0.2 mmol scale. [g]The reaction time was 14 h. [h]The reaction time was 13 h. [i]With 1.5 equiv of Selectfluor. [j]The reaction was conducted with 0.5 mmol of **1p**, 0.50 equiv of $AgNO_3$, 1.20 equiv of $K_2S_2O_8$, and 1.5 equiv of **2a** in 1.2 mL of H_2O and 1.2 mL of DCM at 10 °C for 24 h under an Ar atmosphere. [k]The reaction was conducted with 0.2 mmol of **1p**, 0.50 equiv of $AgNO_3$, 1.20 equiv of $K_2S_2O_8$, and 1.5 equiv of NBS (**2b**) in 0.5 mL of H_2O and 0.5 mL of DCM at 10 °C for 36 h under an Ar atmosphere.

= 90:10; Figure 1). Even the reaction of β-substituted substrate **3n** formed **3na** and **3nb** with high selectivity with respect to **3na′** and **3nb′**.

To demonstrate the synthetic utility and practicality of the method, we conducted the gram-scale reaction of **1a** with NBS under standard conditions, which afforded 1.4464 g of **3ab** in 65% yield. The synthetic potential of this product was also demonstrated (Scheme 2). Product **3ab** could be reduced with

Scheme 2. Representative Synthetic Potential of 3ab[a]

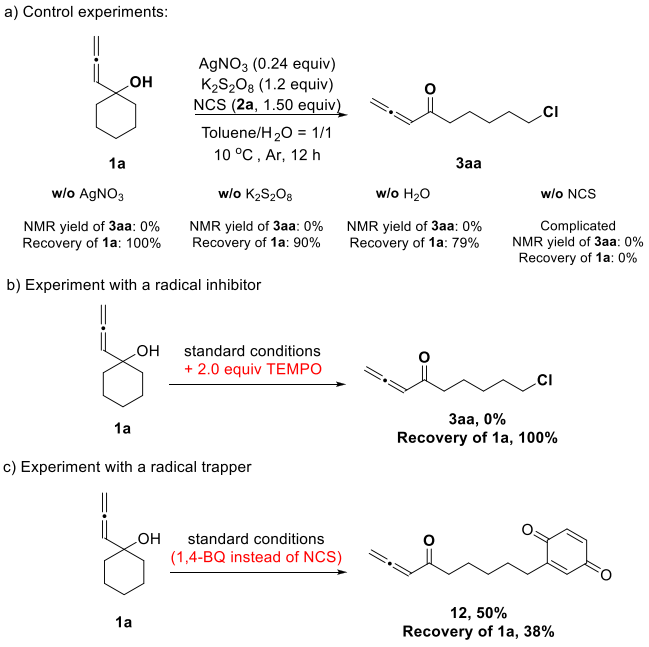


[a]Reaction conditions: (a) **3ab** (0.2 mmol), $LiAlH_4$ (0.5 equiv), Et_2O, 0 °C, then rt, 12 h; (b) **3ab** (0.2 mmol), NaI (5.0 equiv), acetone, 70 °C, 12 h; (c) **3ab** (0.2 mmol), $AgBF_4$ (0.4 equiv), CH_3CN, 100 °C, 12 h; (d) **3ab** (0.2 mmol), LiI (1.2 equiv), AcOH, 25 °C, 3 h; (e) **3ab** (0.24 mmol), aniline (0.2 mmol), Na_2CO_3 (0.1 M), rt, 11 h; (f) **3ab** (0.2 mmol), 4-bromophenol (1.0 equiv), K_2CO_3 (3.0 equiv), acetone, 65 °C, 12 h; (g) **3ab** (0.24 mmol), diethyl malonate (0.2 mmol), K_2CO_3 (0.2 equiv), EtOH, 60 °C, 12 h; (h) **3ab** (0.2 mmol), 1,3-cyclohexanedione (1.0 equiv), Ph_3P (0.1 equiv), toluene, 70 °C, 12 h.

$LiAlH_4$ to provide secondary alcohol **4** in 77% yield.[12] The bromide could be converted to iodide **5** in 95% yield.[6j] The cyclization of **3ab** worked smoothly to provide furan **6** in 70% yield.[13] Alternatively, α-allenyl ketone **3ab** is a very good Michael acceptor for transformations such as 1,4-Michael addition with LiI to generate β,γ-unsaturated enone **7**[14] and 1,4-addition with aniline or 4-bromophenol to provide α,β-unsaturated ketones *Z*-**8** and *E*-**9** in high to excellent yields.[15] **3ab** could also be readily converted to α-pyrone derivative **10** with diethyl malonate.[16] The tandem reaction of **3ab** and 1,3-cyclohexanedione constructed heterocycle **11** in 80% yield via triphenylphosphine-catalyzed umpolung addition and intramolecular conjugate addition.[17]

To gain insight into the mechanism, we carried out some control experiments. First of all, for the ring-opening chlorination reactions, control experiments revealed that the combination of $AgNO_3$, $K_2S_2O_8$, and H_2O was necessary (Scheme 3a). Water is believed to increase the solubility of

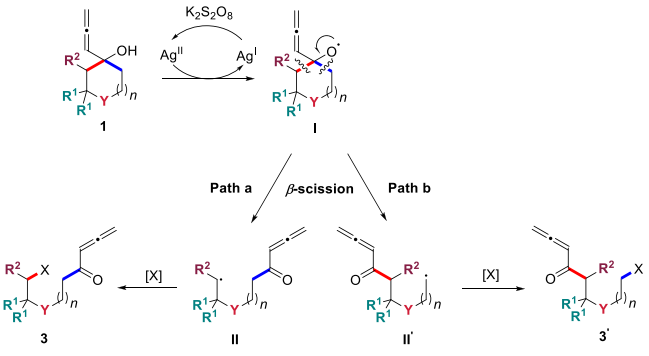
Scheme 3. Mechanistic Studies



$K_2S_2O_8$ in the reaction mixture. Besides, if a capturing agent such as NCS was not added, we did not observe ring expansion products, and the reaction led to a complicated result. We found that the reaction was suppressed when 2.0 equiv of TEMPO was added under the standard conditions (Scheme 3b). In addition, the desired product **12** was obtained when 1,4-benzoquinone was used as a radical trapper (Scheme 3c).[18] These results indicated the involvement of a radical pathway in the reaction.

On the basis of these results, the plausible mechanism shown in Scheme 4 is proposed. First, Ag^I is oxidized by $K_2S_2O_8$ to generate the Ag^{II} species.[19] Subsequently, cycloalkanol **1** is oxidized by the metastable Ag^{II} species to give cycloalkoxy radical intermediate **I**. β-Scission of cycloalkoxy radical intermediate **I** generates two ring-opened alkyl radical intermediates, **II** and **II′**, both of which are then intercepted

Scheme 4. Proposed Mechanism



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by NCS, NBS, or Selectfluor to eventually afford the halogen-substituted allenyl ketones **3** and **3′**, respectively.

In summary, we have developed an efficient procedure for the preparation of halogen-substituted allenyl ketones through silver-catalyzed oxidative ring opening of allenyl cycloalkanols. A wide range of products with substituents at different positions have been synthesized. These allenyl ketones can be easily converted to other useful molecules. Preliminary control experiments indicated a radical pathway. Further studies in this area are being conducted in this laboratory.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c00452.

Experimental details and data (PDF)

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) For a review and example of allenes in natural products and bioactive molecules, see: (a) Hoffmann-Röder, A.; Krause, N. Synthesis and Properties of Allenic Natural Products and Pharmaceuticals. *Angew. Chem., Int. Ed.* **2004**, *43*, 1196. (b) Wan, Z.; Nelson, S. G. Optically Active Allenes from β-Lactone Templates: Asymmetric Total Synthesis of (−)-Malyngolide. *J. Am. Chem. Soc.* **2000**, *122*, 10470.

(2) For selected books, reviews, and examples of nucleophilic addition and cyclization reactions of allenyl ketones, see: (a) Schuster, H. F.; Coppola, G. M. *Allenes in Organic Synthesis*; Wiley: New York, 1984. (b) Ma, S. Transition Metal-Catalyzed/Mediated Reaction of Allenes with a Nucleophilic Functionality Connected to the α-Carbon Atom. *Acc. Chem. Res.* **2003**, *36*, 701. (c) Ye, J.; Ma, S. Palladium-Catalyzed Cyclization Reation of Allenes in the Presence of Unsaturated Carbon-Carbon Bonds. *Acc. Chem. Res.* **2014**, *47*, 989. (d) Fan, X.; He, Y.; Zhang, X. Recent Advances in the Reaction of 1,2-Allenic Ketones and α-Allenic Alcohols. *Chem. Rec.* **2016**, *16*, 1635. (e) Ma, S.; Yu, Z. Efficient Synthesis of 4-(3′-Furanyl)-butenolide Derivatives via Pd^{II}-Catalyzed Oxidative Heterodimeric Cyclization Reaction of 2,3-Allenoic Acids and 1,2-Allenyl Ketones. *Chem. - Eur. J.* **2004**, *10*, 2078. (f) Shi, X.; He, Y.; Zhang, X.; Fan, X. Selective Syntheses of Diversely Substituted 2-Hydroxy-4′-hydroxybenzophenones through [4 + 2] or [3 + 3] Annulation of Penta-3,4-dien-2-ones with 3-Formylchromones. *Org. Chem. Front.* **2017**, *4*, 1967. (g) Wang, Q.; Zhang, T.; Fan, Y.; Fan, X. Synthesis of functionalized cyclopentenes through allenic ketone-based multicomponent reactions. *Org. Biomol. Chem.* **2018**, *16*, 8854. (h) Feng, T.; Tian, M.; Zhang, X.; Fan, X. Tunable Synthesis of Functionalized Cyclohexa-1,3-dienes and 2-Aminobenzophenones/Benzoate from the Cascade Reactions of Allenic Ketones/Allenoate with Amines and Enones. *J. Org. Chem.* **2018**, *83*, 5313. (i) Zhang, T.; Wang, Q. [Bmim]$ZnBr_3$-Promoted Tandem Reaction of Enaminoester and Allenic Ketones for the Synthesis of Substituted Nicotinate Derivatives. *Youji Huaxue* **2018**, *38*, 498. (j) Li, S.; Wu, X.-X.; Chen, S. Base-promoted direct synthesis of functionalized N-arylindoles via the cascade reactions of allenic ketones with indoles. *Org. Biomol. Chem.* **2019**, *17*, 789. (k) Sahoo, S.; Sarkar, D. Revisiting the Addition of in-situ Nucleophiles to Allenic Ketones: An Entry Towards Synthesis of Benzodioxins. *Eur. J. Org. Chem.* **2020**, *2020*, 1727.

(3) For a selected review and examples of the synthesis of allenyl ketones, see: (a) Li, L.; Ma, S. Synthesis and Reaction of 1,2-Allenyl Ketones. *Chin. J. Org. Chem.* **2000**, *20*, 850. (b) Yu, S.; Ma, S. How easy are the syntheses of allenes? *Chem. Commun.* **2011**, *47*, 5384. (c) Alcaide, B.; Almendros, P.; Martínez del Campo, T. Allene Substitution-Controlled Switching of Dimerization to Cycloisomerization in the Pd^{II}-Catalyzed Reaction of Terminal α-Allenones. *Eur. J. Org. Chem.* **2007**, *2007*, 2844. (d) Yu, X.; Ren, H.; Xiao, Y.; Zhang, J. Efficient Assembly of Allenes, 1,3-Dienes, and 4*H*-Pyrans by Catalytic Regioselective Nucleophilic Addition to Electron-Deficient 1,3-Conjugated Enynes. *Chem. - Eur. J.* **2008**, *14*, 8481. (e) Xi, C.; Yan, X.; You, W.; Takahashi, T. Coupling Reactions of Zirconate Complexes Induced by Carbonyl Compounds. *Angew. Chem., Int. Ed.* **2009**, *48*, 8120. (f) Yu, X.; Zhang, J. Regioselective Addition of Organometallic Reagents to 2-(1-Alkynyl)-2-alken-1-ones for an Efficient Synthesis of Substituted 1,2-Allenyl Ketones. *Adv. Synth. Catal.* **2011**, *353*, 1265. (g) Kim, H. Y.; Li, J.-Y.; Oh, K. Studies on Elimination Pathways of β-Halovinyl Ketones Leading to Allenyl and Propargyl Ketones and Furans under the Action of Mild Bases. *J. Org. Chem.* **2012**, *77*, 11132. (h) Yao, Q.; Liao, Y.; Lin, L.; Lin, X.; Ji, J.; Liu, X.; Feng, X. Efficient Synthesis of Chiral Trisubstituted 1,2-Allenyl Ketones by Catalytic Asymmetric Conjugate Addition of Malonic Esters to Enynes. *Angew. Chem., Int. Ed.* **2016**, *55*, 1859.

(4) For examples of oxidation of secondary 2,3-allenols and homopropargylic alcohols to ketones, see: (a) Fan, X.; Qu, Y.; Wang, Y.; Zhang, X.; Wang, J. Ru(III)-catalyzed oxidation of homopropargyl alcohols in ionic liquid: an efficient and green route to 1,2-allenic ketones. *Tetrahedron Lett.* **2010**, *51*, 2123. (b) Zhang, X.; Qu, Y.; Wang, Y.; Fan, X. Synthesis of 1,2-allenic ketones through oxidation of homopropargyl alcohols with CrO_3(cat.)/TBHP under MWI. *Chin. Chem. Lett.* **2011**, *22*, 268. (c) Ma, S.; Liu, J.; Li, S.; Chen, B.; Cheng, J.; Kuang, J.; Liu, Y.; Wan, B.; Wang, Y.; Ye, J.; Yu, Q.; Yuan, W.; Yu, S. Development of a General and Practical Iron Nitrate/TEMPO-Catalyzed Aerobic Oxidation of Alcohols to Aldehydes/Ketones: Catalysis with Table Salt. *Adv. Synth. Catal.* **2011**, *353*, 1005. (d) Gao, S.; Liu, Y.; Ma, S. CuCl-catalyzed aerobic oxidation of 2,3-allenols to 1,2-allenic ketones with 1:1 combination of phenanthroline and bipyridine as ligands. *Beilstein J. Org. Chem.* **2011**, *7*, 396. (e) Liu, J.; Ma, S. Room temperature $Fe(NO_3)_3{\cdot}9H_2O$/TEMPO/NaCl-catalyzed aerobic oxidation of homopropargylic alcohols. *Tetrahedron* **2013**, *69*, 10161.

(5) For selected reviews of activation of unstrained C−C bonds, see: (a) Dermenci, A.; Coe, J. W.; Dong, G. Direct activation of relatively unstrained carbon-carbon bonds in homogeneous systems. *Org. Chem. Front.* **2014**, *1*, 567. (b) Chen, F.; Wang, T.; Jiao, N. Recent Advances in Transition-Metal-Catalyzed Functionalization of Unstrained Carbon-Carbon Bonds. *Chem. Rev.* **2014**, *114*, 8613. (c) Song, F.; Gou, T.; Wang, B.; Shi, Z. Catalytic activations of unstrained C−C

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bond involving organometallic intermediates. *Chem. Soc. Rev.* **2018**, *47*, 7078. (d) Wu, X.; Zhu, C. Recent Advances in Radical-Mediated C−C Bond Fragmentation of Non-Strained Molecules. *Chin. J. Chem.* **2019**, *37*, 171. (e) Lutz, M. D. R.; Morandi, B. Metal-Catalyzed Carbon−Carbon Bond Cleavage of Unstrained Alcohols. *Chem. Rev.* **2021**, *121*, 300.

(6) For selected examples of activation of unstrained C−C bonds, see: (a) Suggs, J. W.; Jun, C.-H. Directed Cleavage of Carbon-Carbon Bonds by Transition Metals: The α-Bonds of Ketones. *J. Am. Chem. Soc.* **1984**, *106*, 3054. (b) Jun, C.-H.; Lee, H. Catalytic Carbon-Carbon Bond Activation of Unstrained Ketone by Soluble Transition-Metal Complex. *J. Am. Chem. Soc.* **1999**, *121*, 880. (c) Xia, Y.; Lu, G.; Liu, P.; Dong, G. Catalytic activation of carbon−carbon bonds in cyclopentanones. *Nature* **2016**, *539*, 546. (d) Xia, Y.; Wang, J.; Dong, G. Distal-Bond-Selective C−C Activation of Ring-Fused Cyclopentanones: An Efficient Access to Spiroindanones. *Angew. Chem., Int. Ed.* **2017**, *56*, 2376. (e) Guo, J.; Hu, A.; Chen, Y.; Sun, J.; Tang, H.; Zuo, Z. Photocatalytic C−C Bond Cleavage and Amination of Cycloalkanols by Cerium(III) Chloride Complex. *Angew. Chem., Int. Ed.* **2016**, *55*, 15319. (f) Chen, Y.; Du, J.; Zuo, Z. Selective C−C Bond Scission of Ketones via Visible-Light-Mediated Cerium Catalysis. *Chem.* **2020**, *6*, 266. (g) Yayla, H.; Wang, H.; Tarantino, K.; Orbe, H.; Knowles, R. Catalytic Ring-Opening of Cyclic Alcohols Enabled by PCET Activation of Strong O-H Bonds. *J. Am. Chem. Soc.* **2016**, *138*, 10794. (h) Zhao, K.; Yamashita, K.; Carpenter, J.; Sherwood, T.; Ewing, W.; Cheng, P.; Knowles, R. Catalytic Ring Expansions of Cyclic Alcohols Enabled by Proton-Coupled Electron Transfer. *J. Am. Chem. Soc.* **2019**, *141*, 8752. (i) Fan, X.; Zhao, H.; Yu, J.; Bao, X.; Zhu, C. Regiospecific synthesis of distally chlorinated ketones via C−C bond cleavage of cycloalkanols. *Org. Chem. Front.* **2016**, *3*, 227. (j) Wang, D.; Mao, J.; Zhu, C. Visible light-promoted ring-opening functionalization of unstrained cycloalkanols *via* inert C−C bond scission. *Chem. Sci.* **2018**, *9*, 5805. (k) Wang, M.; Li, M.; Yang, S.; Xue, X.-S.; Wu, X.; Zhu, C. Radical-mediated C−C cleavage of unstrained cycloketones and DFT study for unusual regioselectivity. *Nat. Commun.* **2020**, *11*, 672. (l) Huang, F.; Xie, J.; Sun, J.; Wang, Y.; Dong, X.; Qi, L.; Zhang, B. Regioselective Synthesis of Carbonyl-Containing Alkyl Chlorides via Silver-Catalyzed Ring-Opening Chlorination of Cycloalkanols. *Org. Lett.* **2016**, *18*, 684. (m) Wang, N.; Gu, Q.; Li, Z.; Li, Z.; Guo, Y.; Guo, Z.; Liu, X. Direct Photocatalytic Synthesis of Medium-Sized Lactams by C−C Bond Cleavage. *Angew. Chem., Int. Ed.* **2018**, *57*, 14225. (n) Huang, L.; Ji, T.; Rueping, M. Remote Nickel-Catalyzed Cross-Coupling Arylation via Proton-Coupled Electron Transfer-Enabled C−C Bond Cleavage. *J. Am. Chem. Soc.* **2020**, *142*, 3532. (o) Shi, J.-L.; Wang, Y.; Wang, Z.; Dou, B.; Wang, J. Ring-opening iodination and bromination of unstrained cycloalkanols through β-scission of alkoxy radicals. *Chem. Commun.* **2020**, *56*, 5002. (p) Ji, T.; Chen, X.-Y.; Huang, L.; Rueping, M. Remote Trifluoromethylthiolation Enabled by Organophotocatalytic C−C Bond Cleavage. *Org. Lett.* **2020**, *22*, 2579.

(7) (a) Pagliaro, M.; Ciriminna, R. New fluorinated functional materials. *J. Mater. Chem.* **2005**, *15*, 4981. (b) Berger, R.; Resnati, G.; Metrangolo, P.; Weber, E.; Hulliger, J. Organic fluorine compounds: a great opportunity for enhanced materials properties. *Chem. Soc. Rev.* **2011**, *40*, 3496.

(8) (a) Smart, B. E. Fluorine substituent effects (on bioactivity). *J. Fluorine Chem.* **2001**, *109*, 3. (b) Shimizu, M.; Hiyama, T. Modern Synthetic Methods for Fluorine-Substituted Target Molecules. *Angew. Chem., Int. Ed.* **2005**, *44*, 214. (c) Bégué, J. P.; Bonnet-Delpon, D. Recent advances (1995−2005) in fluorinated pharmaceuticals based on natural products. *J. Fluorine Chem.* **2006**, *127*, 992. (d) Yerien, D. E.; Bonesi, S.; Postigo, A. Fluorination methods in drug discovery. *Org. Biomol. Chem.* **2016**, *14*, 8398.

(9) O'Hagan, D. Understanding organofluorine chemistry. An introduction to the C-F bond. *Chem. Soc. Rev.* **2008**, *37*, 308.

(10) (a) Patani, G. A.; LaVoie, E. J. Bioisosterism: A Rational Approach in Drug Design. *Chem. Rev.* **1996**, *96*, 3147. (b) Meanwell, N. A. Synopsis of Some Recent Tactical Application of Bioisosteres in Drug Design. *J. Med. Chem.* **2011**, *54*, 2529.

(11) For selected reviews of the formation of C(sp^3)−F bonds, see: (a) Wu, J. Review of recent advances in nucleophilic C-F bond-forming reactions at sp^3 centers. *Tetrahedron Lett.* **2014**, *55*, 4289. (b) Champagne, P. A.; Desroches, J.; Hamel, J.-D.; Vandamme, M.; Paquin, J. F. Monofluorination of Organic Compounds: 10 Years of Innovation. *Chem. Rev.* **2015**, *115*, 9073. (c) Tarantino, G.; Hammond, C. Catalytic C(sp^3)−F bond formation: recent achievements and pertaining challenges. *Green Chem.* **2020**, *22*, 5195.

(12) Westerlund, A.; Gras, J.; Carlson, R. Investigation of 1-bromo-3-buten-2-one as building block in organic synthesis. *Tetrahedron* **2001**, *57*, 5879.

(13) Marshall, J.; Robinson, E. A Mild Method for the Synthesis of Furans. Application to 2,5-Bridged Furano Macrocyclic Compounds. *J. Org. Chem.* **1990**, *55*, 3450.

(14) Shi, Z.; Li, L.; Ma, S. Efficient Synthesis of 4-Halo-4-penten-2-ones and 3-Halo-3-butenoic Acids/Esters *via* Hydrohalogenation Reaction of 3,4-Pentadien-2-one and 2,3-Butadienoic Acid/Methyl Ester. *J. Org. Chem.* **1998**, *63*, 4522.

(15) Tang, X.; Yan, Z.; Chen, W.; Gao, Y.; Mao, S.; Zhang, Y.; Wang, Y. Aza-Michael reaction promoted by aqueous sodium carbonate solution. *Tetrahedron Lett.* **2013**, *54*, 2669.

(16) Yu, S.; Yin, S.; Ma, S. Studies on K_2CO_3-Catalyzed 1,4-Addition of 1,2-Allenic Ketones with Diethyl Malonate: Controlled Selective Synthesis of β,γ-Unsaturated Enones and α-Pyrones. *J. Org. Chem.* **2003**, *68*, 8996.

(17) Lu, C.; Lu, X. Tandem Reactions to Construct Heterocycles via Phosphine-Catalyzed Umpolung Addition and Intramolecular Conjugate Addition. *Org. Lett.* **2002**, *4*, 4677.

(18) (a) Ilangovan, A.; Saravanakumar, S.; Malayappasamy, S. γ-Carbonyl Quinones: Radical Strategy for the Synthesis of Evelynin and Its Analogues by C−H Activation of Quinones Using Cyclopropanols. *Org. Lett.* **2013**, *15*, 4968. (b) Rao, P.; Hayon, E. Experimental determination of the redox potential of the superoxide radical $^{\cdot}O_2^-$. *Biochem. Biophys. Res. Commun.* **1973**, *51*, 468.

(19) For selected reviews of the mechanism involving Ag^{II}, see: (a) Anderson, J. M.; Kochi, J. K. Silver (I)-Catalyzed Oxidative Decarboxylation of Acids by Peroxydisulfate. The Role of Silver (II). *J. Am. Chem. Soc.* **1970**, *92*, 1651. (b) Fang, G.; Cong, X.; Zanoni, G.; Liu, Q.; Bi, X. Silver-Based Radical Reactions: Development and Insights. *Adv. Synth. Catal.* **2017**, *359*, 1422.

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